Chapter 15

Implementation of Social Choice Functions by Mechanisms

In the last chapter, we introduced several key notions: social choice functions, direct mechanisms, and indirect mechanisms. In the current chapter, we explain the process by which a social choice function can be realized through a direct mechanism or an indirect mechanism through illustrative examples. The main example considered is that of a selling agent who wishes to allocate an object to one of two buying agents against a monetary payment. We first define three natural social choice functions SCF1, SCF2, and SCF3, and explore their implementability by direct mechanisms. Next we turn to indirect mechanisms for implementing these social choice functions (first price auction and second price auction) and bring out the idea behind indirect mechanisms. Our discussion of direct and indirect mechanisms leads to the key notions of *dominant strategy implementation* and *Bayesian Nash implementation* followed by the notions of *dominant strategy incentive compatibility* and *Bayesian incentive compatibility*.

15.1 Implementation by Direct Mechanisms

We start with the example on supplier selection from the previous chapter to motivate implementation by direct mechanisms.

Supplier Selection Problem

Example 15.1 (Supplier Selection). Recall Example 14.3 where $N = \{1, 2\}$; $X = \{x, y, z\}$; $\Theta_1 = \{a_1\}$; , $\Theta_2 = \{a_2, b_2\}$ and $\Theta = \{(a_1, a_2), (a_1, b_2)\}$. The payoff functions are reproduced here for ready reference:

$$\begin{aligned} u_1(x, a_1) = 100;\ \ u_1(y, a_1) = 50;\ \ u_1(z, a_1) = 0 \\ u_2(x, a_2) = 0;\ \ u_2(y, a_2) = 50;\ \ u_2(z, a_2) = 100 \\ u_2(x, b_2) = 0;\ \ u_2(y, b_2) = 50;\ \ u_2(z, b_2) = 25. \end{aligned}$$

Suppose the social planner (in this case, the buyer) wishes to implement the social choice function f with $f(a_1, a_2) = y$ and $f(a_1, b_2) = x$. Announcing this as the social choice function, let us say the social planner asks the agents to reveal their types. Agent 1 has nothing to reveal since its type is common knowledge (as his type set is a singleton). We will now check whether agent 2 would be willing to truthfully reveal its type.

- If $\theta_2 = a_2$, then, because $f(a_1, a_2) = y$ and $f(a_1, b_2) = x$ and $u_2(y, a_2) > u_2(x, a_2)$, agent 2 is happy to reveal a_2 as its type.
- However if $\theta_2 = b_2$, then because $u_2(y, b_2) > u_2(x, b_2)$ and $f(a_1, b_2) = x$, agent 2 would wish to lie and claim that its type is a_2 and not b_2.

Thus though the social planner would like to implement an SCF $f(\cdot)$, the social planner would be unable to implement the above SCF since one of the agents (in this case agent 2) does not find it a best response to reveal the true type.

On the other hand, let us say the social planner wishes to implement the social choice function f given by $f(a_1, a_2) = z$ and $f(a_1, b_2) = y$. One can show in this case that the SCF can be implemented. Table 15.1 lists all the nine SCFs and their implementability. □

SCF		Implementable
$f(a_1, a_2)$	$f(a_1, b_2)$	
x	x	$\surd$
x	y	$\times$
x	z	$\times$
y	x	$\times$
y	y	$\surd$
y	z	$\times$
z	x	$\times$
z	y	$\surd$
z	z	$\surd$

Table 15.1: Social choice functions and their implementability

Selling an Indivisible Object

We now consider three social choice functions in the context of the problem of a selling agent (agent 0) seeking to allocate a single indivisible object to one of two buying agents (agents 1 and 2) against a monetary payment (See Figure 15.1). We have already defined two such social choice functions SCF1 and SCF2 in the previous chapter. We recall those functions and also define a third social choice function SCF3. We first consider SCF1. Note that $\Theta = \Theta_0 \times \Theta_1 \times \Theta_2$. The function $f(\theta) = (y_0(\theta), y_1(\theta), y_2(\theta), t_0(\theta), t_1(\theta), t_2(\theta))\ \ \forall \theta = (\theta_0, \theta_1, \theta_2) \in \Theta$ is defined as

$$\begin{aligned} y_0(\theta) &= 0 \\ y_1(\theta) &= 1 \quad \text{if } \theta_1 \geq \theta_2 \\ &= 0 \quad \text{if } \theta_1 < \theta_2 \\ y_2(\theta) &= 1 \quad \text{if } \theta_1 < \theta_2 \\ &= 0 \quad \text{if } \theta_1 \geq \theta_2 \end{aligned}$$

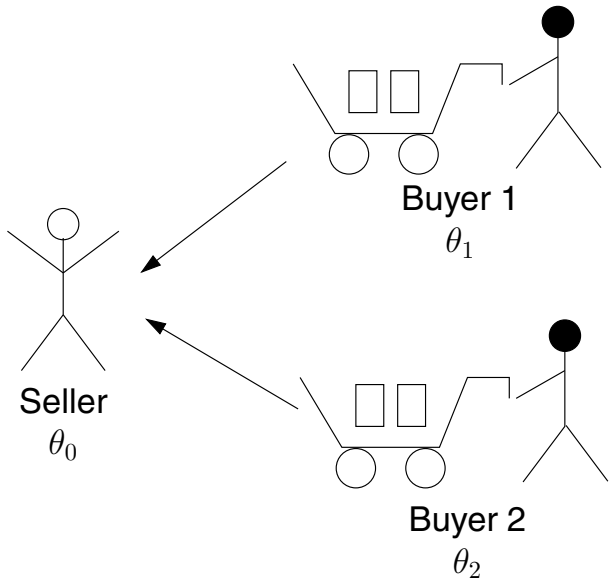

Fig. 15.1: A selling scenario with two buying agents

$$
\begin{aligned}
t_1(\theta) &= -y_1(\theta)\ \theta_1 \\
t_2(\theta) &= -y_2(\theta)\ \theta_2 \\
t_0(\theta) &= -(t_1(\theta) + t_2(\theta)).
\end{aligned}
$$

Note that the object is allocated to whichever buyer has higher valuation (with a tie resolved in favor of agent 1) and the winning buyer has to pay her valuation to the seller. We note that the social choice function is attractive to the seller since the seller will capture all of the benefits that are generated by the object. The social choice function SCF2 is different from SCF1 in that the payment to be made by the winning buyer is equal to the second highest valuation. So, only the following terms will be different in SCF2:

$$
\begin{aligned}
t_1(\theta) &= -y_1(\theta)\ \theta_2 \\
t_2(\theta) &= -y_2(\theta)\ \theta_1
\end{aligned}
$$

We define a third social choice function, SCF3, which has the same allocation rule as SCF1 and SCF2 but the payment by the winning buyer is equal to half her valuation. This is represented as:

$$
\begin{aligned}
t_1(\theta) &= -y_1(\theta)\ \frac{\theta_1}{2} \\
t_2(\theta) &= -y_2(\theta)\ \frac{\theta_2}{2}
\end{aligned}
$$

We now investigate the implementability, through direct mechanisms, of SCF1, SCF2, and SCF3.

Example 15.2 (Implementability of SCF1). Let us assume that θ_1 and θ_2 are drawn independently from a uniform distribution over $[0, 1]$. The following analysis show that SCF1 is not implementable.

Suppose buyer 2 announces his true value θ_2. Let us say the valuation of buyer 1 is θ_1 and he announces $\hat{\theta}_1$. If $\theta_2 \leq \hat{\theta}_1$, then buyer 1 is the winner and his utility will be $\theta_1 - \hat{\theta}_1$. If $\theta_2 > \hat{\theta}_1$, then buyer 2 is the winner and buyer 1's utility is zero. Since buyer 1 wishes to maximize his expected utility, he solves the problem

$$\max_{\hat{\theta}_1}(\theta_1 - \hat{\theta}_1)\ P\{\theta_2 \leq \hat{\theta}_1\}$$

Since θ_2 is uniformly distributed on [0,1],

$$P\{\theta_2 \leq \hat{\theta}_1\} = \hat{\theta}_1$$

Thus buyer 1 tries to solve the problem:

$$\max_{\hat{\theta}_1}(\theta_1 - \hat{\theta}_1)\hat{\theta}_1$$

This problem has the solution

$$\hat{\theta}_1 = \frac{\theta_1}{2}$$

Thus if buyer 2 is truthful, the best response for buyer 1 is to announce $\frac{\theta_1}{2}$.

Similarly if buyer 1 always announces his true valuation θ_1, then the best response of buyer 2 is to announce $\frac{\theta_2}{2}$ when his true type is θ_2. Thus there is no incentive for the buyers to announce their true valuations. A social planner who wishes to realize the above social choice function finds that rational players will not reveal their true private values. Thus SCF1 cannot be implemented. □

Example 15.3 (Implementability of SCF2). We now show that the function SCF2 can be implemented. Suppose θ_i is the true valuation of buyer i ($i = 1, 2$). Let us say buyer 2 announces his valuation as $\hat{\theta}_2$. There are two cases: (1) $\theta_1 \geq \hat{\theta}_2$ and (2) $\theta_1 < \hat{\theta}_2$.

Case 1: $\theta_1 \geq \hat{\theta}_2$

Let $\hat{\theta}_1$ be the announcement of buyer 1. Here there are two cases.

- If $\hat{\theta}_1 \geq \hat{\theta}_2$, then buyer 1 wins and his payoff is $\theta_1 - \hat{\theta}_2 \geq 0$.
- If $\hat{\theta}_1 < \hat{\theta}_2$, then buyer 1 does not win and his payoff is 0.
- Thus in this case, the maximum payoff possible is $\theta_1 - \hat{\theta}_2 \geq 0$.

If $\hat{\theta}_1 = \theta_1$ (that is, buyer 1 announces his true valuation), then payoff for buyer 1 equals $\theta_1 - \hat{\theta}_2$, which happens to be the maximum possible payoff as shown above. Thus announcing θ_1 is a best response for buyer 1 whenever $\theta_1 \geq \hat{\theta}_2$.

Case 2: $\theta_1 < \hat{\theta}_2$

Here again there are two cases: $\hat{\theta}_1 \geq \hat{\theta}_2$ and $\hat{\theta}_1 < \hat{\theta}_2$.

- If $\hat{\theta}_1 \geq \hat{\theta}_2$, then buyer 1 wins and his payoff is $\theta_1 - \hat{\theta}_2$ which is negative.

- If $\hat{\theta}_1 < \hat{\theta}_2$, then buyer 1 does not win and payoff for him is zero.
- Thus in this case, the maximum payoff possible is 0.

If $\hat{\theta}_1 = \theta_1$, payoff for buyer 1 is 0. By announcing $\hat{\theta}_1 = \theta_1$, his true valuation, buyer 1 gets zero payoff, which in this case is a best response.

We can now make the following observations about this example. Reporting his true valuation is optimal for buyer 1 regardless of what buyer 2 reports. Similarly announcing his true valuation is optimal for buyer 2 whatever the announcement of buyer 1 may be. More formally, reporting true type is a weakly dominant strategy for each player. Thus this social choice function can be implemented even though the valuations (types) are private information. □

Example 15.4 (Implementability of SCF3). Again, suppose θ_1 and θ_2 are the true valuations of buyers 1 and 2, respectively. Assume that buyer 2 reports his true type θ_2. Suppose buyer 1 has type θ_1, then his optimal report $\hat{\theta}_1$ is obtained by solving

$$\max_{\hat{\theta}_1} \left(\theta_1 - \frac{\hat{\theta}_1}{2} \right) P\{\theta_2 \leq \hat{\theta}_1\}$$

This is the same as

$$\max_{\hat{\theta}_1} \left(\theta_1 - \frac{\hat{\theta}_1}{2} \right) \hat{\theta}_1$$

This yields $\hat{\theta}_1 = \theta_1$. Thus it is optimal in expectation for buyer 1 to reveal his true private value if buyer 2 reveals his true value. The same situation applies to buyer 2. Thus, a best response for each agent, given that the other agent truthfully reports his true type, is to report his true type. This implies that SCF3 can be implemented. □

Note. It is important to see the difference in the implementability of SCF2 and SCF3. In the case of SCF2, reporting truth is a best response whatever the other agent reports. In the case of SCF3, reporting truth is a best response in expectation over the type reports of the other agent if the other agent also reports truthfully. We say SCF2 is implementable in dominant strategy equilibrium and SCF3 is implementable in Bayesian Nash equilibrium. We wish to emphasize again that SCF1 is not implementable.

15.2 Implementation by Indirect Mechanisms

The examples above have shown us a possible way in which one can try to implement a social choice function. The protocol we followed for implementing the social choice functions was:

- Announce the social choice function $f : \Theta_1 \times \ldots \times \Theta_n \to X$.
- Ask each agent $i = 1, \ldots, n$ to announce his type θ_i.

- Given announcements $(\hat{\theta}_1, \ldots, \hat{\theta}_n)$ by the agents, choose the outcome $x = f(\hat{\theta}_1, \ldots, \hat{\theta}_n) \in X$.

It is natural to call such a method of trying to implement an SCF as a *direct revelation mechanism* or simply a *direct mechanism*. Another approach to implementing a social choice function is through an *indirect way*. Here the mechanism makes the agents interact through an institutional framework in which there are rules governing the actions the agents would be allowed to play and in which there is a way of transforming these actions into an outcome. The design of the mechanism is such that the actions the agents choose will invariably depend on their private values and become the strategies of the players. Auctions provide a common example of indirect mechanisms and we illustrate below two commonly used auctions, first price auction (FPA) and second price auction (SPA).

Example 15.5 (First Price Sealed Bid Auction for Selling). Consider the seller (agent 0) and two potential buyers (agents $1, 2$) as before. The seller plays the role of an auctioneer and the buyers play the role of bidders in the auction. Each buyer is asked to submit a sealed bid, $b_i \geq 0$ $(i = 1, 2)$. The sealed bids are examined and the buyer with the higher bid is declared the winner. If there is a tie, assume that buyer 1 is declared the winner. The winning buyer pays to the seller an amount equal to his bid. The losing bidder does not pay anything.

We emphasize the subtle difference between the situations in Example 15.2 and the current example. In Example 15.2 (direct mechanism), each buyer is asked to announce (directly) his type (valuation for the object), whereas in the current example 15.5 (indirect mechanism), each buyer is asked to submit a bid for the object. The intent is that the bid submitted will depend on the type. Based on the type, the buyer has a strategy for bidding. So it becomes a game and in an equilibrium of the game, the strategies, hopefully, will indirectly reveal the types of the players.

Let us make the following assumptions:

(1) θ_1, θ_2 are independently drawn from the uniform distribution on $[0, 1]$.
(2) The sealed bid of buyer i takes the form $b_i(\theta_i) = \alpha_i \theta_i$, where $\alpha_i \in (0, 1]$. That the bid will be of the above form is known to both the bidders. The values of α_1, α_2 are not known however and the bidders have to compute these in order to determine their best response bids.

Buyer 1's problem is now to bid in a way so as to maximize his payoff:

$$\max_{b_1 \geq 0} \; (\theta_1 - b_1) \, P\{b_2(\theta_2) \leq b_1\}$$

Since the bid of player 2 is $b_2(\theta_2) = \alpha_2 \theta_2$ and $\theta_2 \in [0, 1]$, the maximum bid of buyer 2 is α_2. Buyer 1 knows this and therefore $b_1 \in [0, \alpha_2]$. Also,

$$\begin{aligned} P\{b_2(\theta_2) \leq b_1\} &= P\{\alpha_2 \theta_2 \leq b_1\} \\ &= P\left\{\theta_2 \leq \frac{b_1}{\alpha_2}\right\} \\ &= \frac{b_1}{\alpha_2} \text{ since } \theta_2 \text{ is uniform over } [0, 1] \end{aligned}$$

Thus buyer 1's problem is:

$$\max_{b_1 \in [0,\alpha_2]} (\theta_1 - b_1)\frac{b_1}{\alpha_2}$$

The solution to this problem is

$$b_1(\theta_1) = \begin{cases} \frac{\theta_1}{2} \text{ if } \frac{\theta_1}{2} \leq \alpha_2 \\ \alpha_2 \text{ if } \frac{\theta_1}{2} > \alpha_2 \end{cases}$$

We can show on similar lines that

$$b_2(\theta_2) = \begin{cases} \frac{\theta_2}{2} \text{ if } \frac{\theta_2}{2} \leq \alpha_1 \\ \alpha_1 \text{ if } \frac{\theta_2}{2} > \alpha_1 \end{cases}$$

Let $\alpha_1 = \alpha_2 = \frac{1}{2}$. Then we get

$$b_1(\theta_1) = \frac{\theta_1}{2} \quad \forall \theta_1 \in \Theta_1 = [0,1]$$
$$b_2(\theta_2) = \frac{\theta_2}{2} \quad \forall \theta_2 \in \Theta_2 = [0,1]$$

Note that if $b_2(\theta_2) = \frac{\theta_2}{2}$, we have $\alpha_2 = \frac{1}{2}$ and $\frac{\theta_1}{2} \leq \alpha_2$ and therefore the best response of buyer 1 is $b_1(\theta_1) = \frac{\theta_1}{2}$. Similarly if $b_1(\theta_1) = \frac{\theta_1}{2}$, we have $\alpha_1 = \frac{1}{2}$ and $\frac{\theta_2}{2} \leq \alpha_1$ and therefore the best response of buyer 2 is $b_2(\theta_2) = \frac{\theta_2}{2}$. Hence the profile $\left(\frac{\theta_1}{2}, \frac{\theta_2}{2}\right)$ is a Bayesian Nash equilibrium of an underlying Bayesian game. This equilibrium can be computed by the rational and intelligent bidders.

This means there is a Bayesian Nash equilibrium of an underlying Bayesian game (induced by the indirect mechanism namely the first price sealed bid auction) that yields the outcome $f(\theta) = (y_0(\theta), y_1(\theta), y_2(\theta), t_0(\theta), t_1(\theta), t_2(\theta))$ such that $\forall \theta = (\theta_0, \theta_1, \theta_2) \in \Theta$,

$$\begin{aligned} y_0(\theta) &= 0 \\ y_1(\theta) &= 1 \qquad \text{if } \theta_1 \geq \theta_2 \\ &= 0 \qquad \theta_1 < \theta_2 \\ y_2(\theta) &= 1 \qquad \text{if } \theta_1 < \theta_2 \\ &= 0 \qquad \theta_1 \geq \theta_2 \\ t_1(\theta) &= -y_1(\theta)\frac{\theta_1}{2} \\ t_2(\theta) &= -y_2(\theta)\frac{\theta_2}{2} \\ t_0(\theta) &= -(t_1(\theta) + t_2(\theta)) \end{aligned}$$

Note that the social choice function that is realized above is SCF3. We emphasize that it is neither SCF1 nor SCF2. □

Example 15.6 (Second Price Sealed Bid Auction). In this indirect mechanism, each buyer is asked to submit a sealed bid $b_i \geq 0$. The bids are examined and the buyer with higher bid is declared the winner. In case there is a tie, buyer 1 is declared the winner.

The winning buyer will pay to the seller an amount equal to the second highest bid. The losing bidder does not pay anything.

In this case, we can show that $b_i(\theta_i) = \theta_i$ for $i = 1, 2$ constitutes a weakly dominant strategy for each player. The arguments are identical to those in Example 15.3.

Thus the game induced by the indirect mechanism second price sealed bid auction has a weakly dominant strategy in which the social choice function SCF2 is implemented. □

Summary of Examples

We can summarize the main points of the current chapter so far as follows.

- The social choice function SCF1 cannot be implemented by a direct mechanism. We still have not investigated if SCF1 can perhaps be implemented by an indirect mechanism.
- The function SCF2 can be implemented in dominant strategies by a direct mechanism. Also, the indirect mechanism, namely second price auction implements SCF2 in dominant strategies.
- The function SCF3 is implemented in Bayesian Nash equilibrium by a direct mechanism. Also the indirect mechanism, namely first price auction, implements it in Bayesian Nash equilibrium.

15.3 Bayesian Game Induced by a Mechanism

Recall that a mechanism is an institution or a framework with a set of rules that prescribe the actions available to players and specify how these action profiles are transformed into outcomes. A mechanism specifies an action set for each player. The outcome function gives the rule for obtaining outcomes from action profiles. Given:

(1) a set of agents $N = \{1, 2, \ldots, n\}$,
(2) type sets $\Theta_1, \ldots, \Theta_n$,
(3) a common prior $\mathbb{P} \in \Delta(\Theta)$ and belief functions $p_i : \Theta_i \to \Delta(\Theta_{-i})$ $(i = 1, \ldots, n)$ that can be derived from $\mathbb{P}$,
(4) a set of outcomes X,
(5) utility functions $u_1, \ldots, u_n$, with $u_i : X \times \Theta_i \to \mathbb{R}$,

a mechanism $\mathscr{M} = (S_1, \ \ldots, \ S_n, g(\cdot))$ induces among the players, a Bayesian game $\langle N, (\Theta_i), (S_i), (p_i), (U_i)\rangle$ where

$$U_i(\theta_1, \ \ldots, \ \theta_n, s_1, \ \ldots, \ s_n) = u_i(g(s_1, \ \ldots, \ s_n), \theta_i).$$

We will use the symbol u_i instead of U_i whenever there is no confusion.

Strategies in the Induced Bayesian Game

For $i = 1, 2, \ldots, n$, a strategy s_i for agent i in the induced Bayesian game is a mapping $s_i : \Theta_i \to S_i$. Thus, given a type $\theta_i \in \Theta_i$, $s_i(\theta_i)$ will give the action of player i corresponding to θ_i. The strategy $s_i(.)$ will specify actions corresponding to types. In the auction scenario, the bid b_i of player i is a function of the valuation θ_i. For example, $b_i(\theta_i) = \alpha_i \theta_i$ is a particular strategy for player i.

Example 15.7 (Bayesian Game Induced by First Price Auction).
First, note that $N = \{0, 1, 2\}$. The type sets are $\Theta_0, \Theta_1, \Theta_2$, and the common prior is some known distribution $\mathbb{P} \in \Delta(\Theta)$. The set of outcomes is

$$X = \{(y_0, y_1, y_2, t_0, t_1, t_2) : y_i \in \{0, 1\}, y_0 + y_1 + y_2 = 1, t_i \in \mathbb{R}\ \forall i \in N\}$$

The utility functions for the buyers 1 and 2 are given by

$$u_i((y_0, y_1, y_2, t_0, t_1, t_2), \theta_i) = y_i \theta_i + t_i; \quad i = 1, 2$$

The strategy sets are given by

$$S_0 = \{\theta_0\};\ S_1 = \mathbb{R}_+\ ;\ \ S_2 = \mathbb{R}_+$$

Note that θ_0 is some known valuation the seller has for the object. Since S_0 is a singleton, the bid of the seller is common knowledge. We include it in the following only for the sake of completeness. The outcome rule $g(\cdot)$ for bid profiles $b = (b_0, b_1, b_2) \in S_0 \times S_1 \times S_2$ is given by

$$g(b) = (y_0(b), y_1(b), y_2(b), t_0(b), t_1(b), t_2(b))$$

such that $\forall (b_0, b_1, b_2) \in S_0 \times S_1 \times S_2$,

$$\begin{aligned}
y_0(b_0, b_1, b_2) &= 0 \\
y_1(b_0, b_1, b_2) &= 1 \qquad \text{if } b_1 \geq b_2 \\
&= 0 \qquad \text{if } b_1 < b_2 \\
y_2(b_0, b_1, b_2) &= 1 \qquad \text{if } b_1 < b_2 \\
&= 0 \qquad \text{if } b_1 \geq b_2 \\
t_1(b_0, b_1, b_2) &= -y_1(b_0, b_1, b_2)\ b_1 \\
t_2(b_0, b_1, b_2) &= -y_2(b_0, b_1, b_2)\ b_2 \\
t_0(b_0, b_1, b_2) &= -(t_1(b_0, b_1, b_2) + t_2(b_0, b_1, b_2)).
\end{aligned}$$

If we specify the common prior $\mathbb{P}$ and the belief functions, it would complete the specification of the Bayesian game induced by the first price auction. □

15.4 Implementation of a Social Choice Function by a Mechanism

We now formalize the notion of implementation of a social choice function by a mechanism.

Definition 15.1 (Implementation of an SCF). *We say a mechanism* $\mathscr{M} = ((S_i)_{i \in N}, g(\cdot))$*, where* $g : S_1 \times \ldots \times S_n \to X$*, implements the social choice function* $f(\cdot)$ *if there is a pure strategy equilibrium* $s^*(\cdot) = (s_1^*(\cdot), \ldots, s_n^*(\cdot))$ *of the Bayesian game* Γ *induced by* $\mathscr{M}$ *such that* $g\left(s_1^*(\theta_1), \ldots, s_n^*(\theta_n)\right) = f\left(\theta_1, \ldots, \theta_n\right), \forall \left(\theta_1, \ldots, \theta_n\right) \in \Theta$.

Figure 15.2 depicts all the building blocks involved in the implementation of a social choice function by an indirect mechanism.

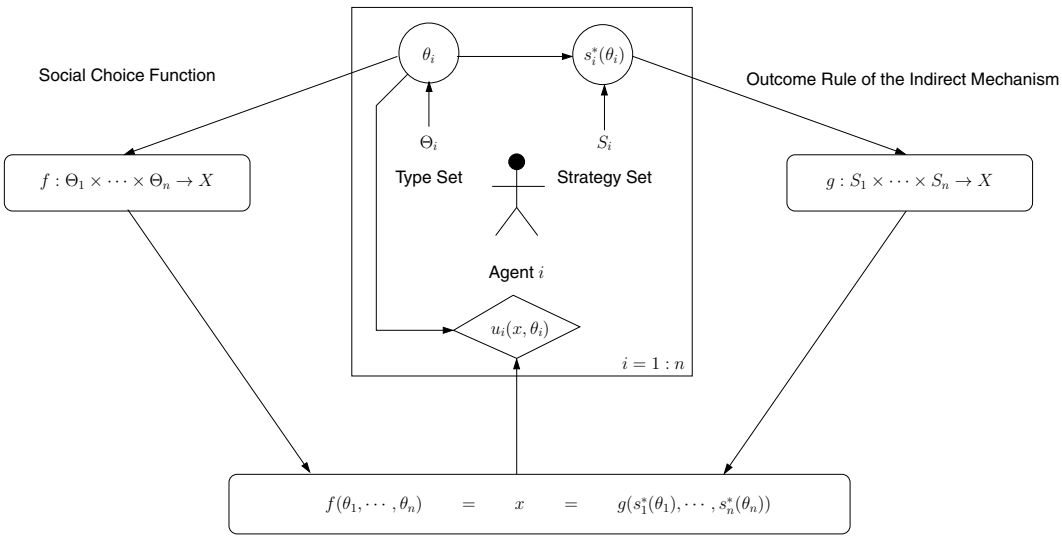

Fig. 15.2: The idea behind implementation by an indirect mechanism

Note that a direct mechanism corresponding to a social choice function $f(\cdot)$ is a special case of an indirect mechanism with the strategy sets same as the type sets and the outcome rule $g(\cdot)$ same as the social choice function $f(\cdot)$. It goes without saying that a Bayesian game is induced by a direct mechanism as well.

Depending on the nature of the underlying equilibrium, two ways of implementing an SCF $f(\cdot)$ are standard in the literature.

Definition 15.2 (Implementation in Dominant Strategies). *We say a mechanism* $\mathscr{M} = ((S_i)_{i \in N}, g(\cdot))$ *implements the social choice function* $f(\cdot)$ *in dominant strategy equilibrium if there is a weakly dominant strategy equilibrium* $s^*(\cdot) = (s_1^*(\cdot), \ldots, s_n^*(\cdot))$ *of the Bayesian game* Γ *induced by* $\mathscr{M}$ *such that*

$$g\left(s_1^*(\theta_1), \ldots, s_n^*(\theta_n)\right) = f\left(\theta_1, \ldots, \theta_n\right) \ \forall \left(\theta_1, \ldots, \theta_n\right) \in \Theta.$$

Since a strongly dominant strategy equilibrium is automatically a weakly dominant strategy equilibrium, the above definition applies to the strongly dominant case also. In the latter case, we could say the implementation is in strongly dominant strategy

equilibrium. The definition also applies to very weakly dominant strategy equilibria and often times this is the implementation that is considered.

Note that the second price auction implements SCF2 in weakly dominant strategy equilibrium. We have also shown that SCF2 can be implemented through a direct mechanism in (weakly) dominant strategies. The latter statement is technically stated by saying that SCF2 is dominant strategy incentive compatible (more about this in the next chapter).

Definition 15.3 (Implementation in Bayesian Nash Equilibrium). *We say that a mechanism* $\mathscr{M} = ((S_i)_{i \in N}, g(\cdot))$ *implements the social choice function* $f(\cdot)$ *in Bayesian Nash equilibrium if there is a pure strategy Bayesian Nash equilibrium* $s^*(\cdot) = (s_1^*(\cdot), \ldots, s_n^*(\cdot))$ *of the game* Γ *induced by* $\mathscr{M}$ *such that*

$$g\left(s_1^*(\theta_1), \ldots, s_n^*(\theta_n)\right) = f\left(\theta_1, \ldots, \theta_n\right) \ \forall \left(\theta_1, \ldots, \theta_n\right) \in \Theta.$$

Note that the first price auction implements SCF3 in Bayesian Nash equilibrium. We have also shown that SCF3 can be implemented through a direct mechanism in Bayesian Nash equilibrium. The latter statement is technically stated by saying that SCF3 is Bayesian incentive compatible (more about this in the next chapter).

Finally, note that SCF1 cannot be implemented using a direct mechanism. In fact, as shown in the next chapter, there is no indirect mechanism that can implement SCF1.

Key Facts about Implementation

Note the requirement on existence of a pure strategy Bayesian Nash equilibrium for the game Γ induced by $\mathscr{M}$. Naturally we must be wary of the possibility that such an equilibrium may not exist.

On the other hand, the game Γ may have more than one equilibrium, but the above definition requires only that *one of them* induces outcomes in accordance with the SCF $f(\cdot)$. Implicitly, then, the above definition assumes that, if multiple equilibria exist, the agents will play the equilibrium that the mechanism designer (social planner) wants.

Another implicit assumption of the above definition is that the game induced by the mechanism is a simultaneous move game, that is all the agents, after learning their types, choose their actions simultaneously.

We have talked about two broad categories of implementations: dominant strategy implementation (DSI) and Bayesian Nash implementation (BNI). It is easy to see that DSI implies BNI but BNI need not imply DSI. Apart from these two broad categories, we can consider other solution concepts for games and define implementability in those equilibria. Examples include: Nash equilibrium (complete information), subgame perfect equilibrium, etc. Chapter 24 treats implementation in Nash equilibrium in detail. At this point, we only state that implementation in

Nash equilibrium is stronger than BNI but weaker than DSI.

The *revelation theorem* which is a fundamental result in mechanism design shows that direct mechanisms and indirect mechanisms are equivalent. Indirect mechanisms (for example, auctions) are appealing and useful because they provide us with practical ways of implementing social choice functions. Direct mechanisms provide an abstraction which is valuable in developing the theory of mechanism design. The revelation theorem is covered in detail in the next chapter.

15.5 Summary and References

In this chapter, we have investigated the process of implementing a social choice function through a direct mechanism or an indirect mechanism. The following are the salient points to be noted.

- A direct mechanism for a social choice function works by trying to extract the private information (types) from the agents truthfully. An indirect mechanism (such as an auction where agents bid) tries to extract truthful information by inducing the agents to play certain strategies that indirectly capture the private information.
- Not all social choice functions can be implemented by direct mechanisms, as shown by the social choice function SCF1.
- Social choice functions such as SCF2 can be implemented by a direct mechanism in dominant strategies.
- Social choice functions such as SCF3 cannot be implemented by a direct mechanism in dominant strategies but may be implemented by a direct mechanism in Bayesian Nash equilibrium.
- A direct mechanism (corresponding to an SCF) induces a Bayesian game among the agents and the induced Bayesian game may have a dominant strategy equilibrium or a Bayesian Nash equilibrium or none. If reporting true types is an equilibrium, then we say the direct mechanism implements the SCF and we say the SCF is incentive compatible.
- Given an SCF, an indirect mechanism is said to implement the SCF if the mechanism induces a Bayesian game having an equilibrium that produces precisely the outcomes of the SCF through the mechanism's outcome rule.

In the next chapter, we show that direct mechanisms and indirect mechanisms are in fact equivalent. This is the central idea of the revelation theorem. The next chapter also defines the key notion of incentive compatibility.

Chapter 24 introduces the notion of implementation in Nash equilibrium in the complete information setting. The chapter also talks about the notion of *strong implementation* which means that the outcomes of the social choice function are reproduced in every equilibrium of the induced Bayesian game.

The contents of this chapter, especially SCF1, SCF2, and SCF3, are based on the examples discussed in the book by Mas-Colell, Whinston, and Green [1]. The procurement versions (buying scenarios) of SCF1, SCF2, and SCF3 are discussed in detail in the monograph by Narahari, Garg, Narayanam, and Prakash [2].

References

[1] Andreu Mas-Colell, Michael D. Whinston, and Jerry R. Green. *Microeconomic Theory*. Oxford University Press, 1995.

[2] Y. Narahari, Dinesh Garg, Ramasuri Narayanam, and Hastagiri Prakash. *Game Theoretic Problems in Network Economics and Mechanism Design Solutions*. Springer, London, 2009.

15.6 Exercises

(1) Investigate the implementability of the eight social choice functions (supplier selection problem) presented in Table 15.1.

(2) Think about what would be an indirect mechanism for implementing social choice functions of the type presented for the supplier selection problem.

(3) Consider a buying scenario with one buying agent (agent 0) and two selling agents (agents $1, 2$). Modify the social choice functions SCF1, SCF2, SCF3 to represent the buying scenario and call the new social choice functions BUY1, BUY2, and BUY3 respectively. Write down these three new social choice functions in the usual notation.

(4) Show that the social choice function BUY1 cannot be implemented by a direct mechanism.

(5) Show that BUY2 and BUY3 can be implemented by appropriate direct mechanisms.

(6) Show that the mechanism *lowest price procurement auction* implements BUY3 in Bayesian Nash equilibrium (Hint: see [2]).

(7) Show that BUY2 is implementable in dominant strategy equilibrium by the mechanism *second lowest price procurement auction*.

(8) In some of the examples discussed in this chapter, it is assumed that a tie between agent 1 and agent 2 will be resolved in favor of agent1. Investigate what would happen in these examples if the tie is resolved in favor of agent 1 or agent 2 using a Bernoulli random variable.

(9) Consider two agents 1 and 2 where agent 1 is the seller of an indivisible item and agent 2 is a prospective buyer of the item. An outcome here is of the form $x = (y_1, y_2, t_1, t_2)$ where $y_i = 1$ if agent i gets the item and t_i denotes the payment received by agent i $(i = 1, 2)$. A natural set of outcomes here is

$$X = \{(y_1, y_2, t_1, t_2) : \ y_1 + y_2 = 1; \ y_1, y_2 \in \{0, 1\}\}.$$

Define the utilities of the agents i $(i = 1, 2)$ appropriately. The type θ_1 of agent

1 (seller) can be interpreted as the willingness to sell of the agent (minimum price at which agent 1 is willing to sell). The type θ_2 of agent 2 (buyer) has the natural interpretation of willingness to pay (maximum price the buyer is willing to pay). Assume that $\Theta_1 = \Theta_2 = [0, 1]$ and that each agent thinks that the type of the other agent is uniformly distributed over the real interval $[0, 1]$. Define the social choice function $f(\theta) = (y_1(\theta), y_2(\theta), t_1(\theta), t_2(\theta))$ by

$$\begin{aligned}
y_1(\theta_1, \theta_2) &= 1 \quad \theta_1 > \theta_2 \\
&= 0 \quad \theta_1 \leq \theta_2 \\
y_2(\theta_1, \theta_2) &= 1 \quad \theta_1 \leq \theta_2 \\
&= 0 \quad \theta_1 > \theta_2 \\
t_1(\theta_1, \theta_2) &= y_2(\theta_1, \theta_2) \left(\frac{\theta_1 + \theta_2}{2} \right) \\
t_2(\theta_1, \theta_2) &= -y_2(\theta_1, \theta_2) \left(\frac{\theta_1 + \theta_2}{2} \right).
\end{aligned}$$

Now define the following indirect mechanism. The seller and the buyer are asked to submit their bids b_1 and b_2 respectively. Trade happens if $b_1 \leq b_2$ and trade does not happen otherwise. If trade happens, the buyer gets the item and pays the seller an amount $\frac{(b_1+b_2)}{2}$. Will this indirect mechanism implement the above social choice function?